Internship Final Report

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University: SVKM's Shri Bhagubhai Mafatlal Polytechnic and College of Engineering

Major: Computer Science (Artificial Intelligence & Machine Learning)

Internship Duration: April 1st, 2026 – April 30th, 2026

Company: ShadowFox

Domain: Artificial Intelligence & Machine Learning

Mentor: Mr. Hariharan

Coordinator: Mr. Aakash

Objectives

The primary objectives of this internship were:

- To understand core concepts of **Artificial Intelligence and Machine Learning**
- To gain hands-on experience in building **ML models for real-world prediction problems**
- To learn the complete **Machine Learning pipeline** from data preprocessing to deployment
- To develop skills in **supervised learning algorithms (regression models)**
- To deploy machine learning models using a web framework
- To strengthen practical implementation skills in Python for AI/ML applications

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Tasks and Responsibilities

During my internship, I worked on two major machine learning projects, focusing on real-world prediction systems:

- **Housing Price Prediction Model:** Developed a regression-based model to predict housing prices by analyzing features such as location, size, and other relevant factors. This involved data preprocessing, handling missing values, and understanding feature relationships through exploratory analysis.
- **Car Price Prediction Model:** Built a machine learning model to estimate car prices using attributes such as present price, kilometers driven, fuel type, and transmission. Applied feature engineering and encoding techniques to convert categorical data into numerical format.
- **Model Training and Evaluation:** Trained regression models and evaluated their performance using appropriate metrics to ensure reliable predictions.
- **Model Deployment using Flask:** Developed a web-based interface where users can input car details and receive predicted prices in real time.
- **Backend Implementation:** Integrated the trained model with application logic using Pandas and NumPy, ensuring proper input structuring and prediction output .

Learning Outcomes

- **Understanding of AI/ML Concepts:** Developed a clear understanding of supervised learning, especially regression techniques used for prediction tasks.
- **End-to-End Model Development:** Learned the complete workflow from data preprocessing to model building, evaluation, and deployment.
- **Practical Implementation Skills:** Gained experience in building real-world projects and converting machine learning models into functional applications.
- **Improved Analytical Thinking:** Enhanced ability to analyze datasets, select relevant features, and improve model performance.

Challenges and Solutions

- **Data Quality Issues:** Faced challenges with missing and inconsistent data in datasets, which were resolved through proper preprocessing and cleaning techniques.
- **Handling Categorical Features:** Converting non-numeric data such as fuel type and transmission required encoding methods to make them suitable for model training.
- **Model Optimization:** Achieving accurate predictions required testing different approaches and refining models based on performance metrics.
- **Deployment Integration:** Ensured that user input from the web interface matched the model's expected input format, resolving errors during prediction.

Conclusion

This internship in the domain of **Artificial Intelligence and Machine Learning** provided valuable hands-on experience in building and deploying ML models.

Through projects like housing and car price prediction, I gained practical knowledge of:

- Machine learning workflows
- Model training and evaluation
- Real-world AI/ML applications
- Deployment using web technologies

This experience has strengthened my foundation in AI/ML and prepared me for advanced development and research in this field. It has also motivated me to further explore advanced machine learning techniques and real-world AI applications.

Acknowledgments

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I express my sincere gratitude to ShadowFox for providing this opportunity to work in the field of Artificial Intelligence and Machine Learning. I would like to thank my mentor, Mr. Hariharan, and my coordinator, Mr. Aakash, for their continuous guidance and support. I also thank my university for encouraging practical learning and helping me grow professionally.